MECHATROLIX
INNOVATE. AUTOMATE. ELEVATE

COMPANY PROFILE

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End-to-end industrial automation services, from certified functional safety design to full plant commissioning, delivered with technical precision across every layer of the production stack.

ABOUT MECHATROLIX

Mechatrolix is a specialist industrial automation consultancy offering integrated EC&I, PLC programming, SCADA development, Robot programming, and functional safety services to manufacturers, process plants, and OEMs across Southern Africa and beyond.

We bridge the gap between engineering design and operational reality, delivering compliant, robust, and maintainable automation solutions that meet international safety standards from concept through commissioning.

Our work spans the full automation lifecycle: electrical specification and schematic design, controller programming, supervisory system integration, robot programming, and project management, under a single accountable engineering partner.

LEADERSHIP

Managing Director

Lithalethu Yotsi
BEng Mechatronics
CMSE - Certified Machinery Safety Expert

OUR SERVICES

01	**Machine Risk Assessment & Functional Safety** Certified machine risk assessments and functional safety design per ISO 13849, IEC 62061, and IEC 61511. Includes SISTEMA calculations, safety concept documentation, and SIL/PLr verification.
02	**PLC/DCS Programming** Controller programming across Siemens TIA Portal (S7-300/400/1200/1500)/PCS7, Allen-Bradley Studio 5000 (ControlLogix/CompactLogix), Beckhoff TwinCAT 3, and OMRON Sysmac. Structured, maintainable code to IEC 61131-3 standards.

03

SCADA / HMI Development & SQL Integration

Full SCADA and HMI solution design and development in WinCC v7.x/v8.x, WinCC Unified/Professional, AVEVA Edge and System Platform, FactoryTalk View SE, and Ignition. Includes historian configuration and SQL Server integration.

04

Low Voltage EC&I Solution Specification

Specification of LV electrical, control, and instrumentation solutions including MCC design, panel specifications, field instrument datasheets, and cable schedules in compliance with applicable LV standards.

05

IEC Electrical Schematics

Complete IEC-compliant electrical schematic packages produced in EPLAN Electric P8 and AutoCAD Electrical, including single line diagrams, loop diagrams, panel layouts, and terminal schedules.

06

Custom .NET Shop Floor Applications

Bespoke Windows-based applications for the shop floor layer: production tracking, operator dashboards, MES interfaces, downtime management, and ERP integration using the .NET framework and SQL Server.

07

FANUC Robot Programming

Programming and integration of FANUC industrial robots using KAREL and Teach Pendant (TP) language for welding, material handling, palletising, and assembly applications on R-30iA/R-30iB controllers.

08

VSD Sizing & Commissioning

Variable speed drive selection, sizing, parameter configuration, and site commissioning across major platforms including Siemens SINAMICS, ABB ACS, Schneider Altivar, and Danfoss VLT series.

09

Machine & Plant Commissioning / Project Management

Full project lifecycle support from FAT and SAT planning through site installation, loop checking, and handover documentation. Automation project management including scheduling, subcontractor coordination, and commissioning sign-off.

STANDARDS AND PLATFORMS

Mechatrolix operates in compliance with and to the requirements of the following international standards and platforms:

Safety Standards	ISO 13849-1/2 • IEC 62061 • IEC 61511 • IEC 60364 • EN ISO 12100
PLC/DCS Platforms	Siemens S7 / TIA Portal • Siemens PCS7• Allen-Bradley Studio 5000 • Beckhoff TwinCAT 3 • OMRON Sysmac NJ/NX • IEC 61131-3
SCADA / HMI	WinCC v7/v8 Unified • AVEVA Edge • AVEVA System Platform • FactoryTalk View SE • Ignition (Inductive Automation)
Schematic Design	EPLAN Electric P8 • AutoCAD Electrical
Networks & Protocols	Profinet • EtherNet/IP • OPC-UA • Modbus TCP • ASi / DeviceNet • Profibus • EtherCAT
Development	.NET Framework / C# • SQL Server • FANUC TP / KAREL • SINAMICS / ACS / Altivar • Python

INDUSTRIES

Our Service are focused on but not limited to the folllowing Industries

Primary	Energy & Utilities	Heavy	Other	Other
Manufacturing	Oil & Gas	Rubber and Plastics	Irrigation ad Agricultural Automation	Warehousing and Distribution Centers
Automotive	Power Generation & Tansmission	Glass Manufacturing	Grain Handling and Silos	Conveyance and Sortation
Mining and Mineral/Chemical Processing	Water and Wastewater Treatment	Cement and Construction Materials	Greenhouse Automation	
Food & Beverage Processing	Coal Handling and Preparation	Steel and Metallurgy	Marine Applications	

WHY MECHATROLIX

- Single-partner accountability: We cover safety, electrical, PLC, SCADA, robotics, and commissioning under one roof , eliminating interface risk between disciplines.
- Standards-compliant by design: All functional safety work is conducted and documented in accordance with ISO 13849, IEC 62061, and IEC 61511 from the outset, not retrofitted.
- Platform-agnostic capability: We program and integrate across Siemens, Allen-Bradley, Beckhoff, and OMRON controllers, with no vendor bias, the right tool for each project .
- Field to enterprise integration: From field instrumentation through PLC logic, SCADA visualisation, SQL historian, and shop floor .NET applications, we deliver full vertical integration.
- Project delivery focus: Our team manages automation projects end-to-end: specification, design, FAT, site installation, loop checking, and commissioning sign-off.

CONTACT US:

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PROJECT PORTFOLIO

Uranium Plant PLC/SCADA/VSD and AS-i protocol commissioning

1. PLC Programming and Control Architecture

PLC programming, hardware configuration, and commissioning of the plant's central control system based on the Siemens S7-1500 PLC platform.

- Developed comprehensive control logic using a hybrid of STL (Statement List), SCL (Structured Control Language), LAD (Ladder Logic), and GRAPH (Sequential Function Chart) programming languages to achieve modular and transparent process control.
- Implemented modular function blocks and diagnostic routines to standardize interlocking, drive control, analog scaling, and alarm management across plant areas.
- Designed control strategies for 34 variable speed drives (VSDs) and 42 direct-on-line (DOL) motor starters, ensuring each device was fully interlocked with field instrumentation, permissive signals, and safety circuits.
- Created sequential control logic for multi-stage process units including the ore scrubbing, pre-thickening, leaching, elution, resin-in-pulp, and precipitation circuits, ensuring correct interdependencies and safe state transitions.

2. VSD Parameterization and Integration

The plant's motor control system utilized Nidec Control Techniques C200 and C300 variable speed drives.

- Carried out parameterization of all VSDs, including motor data entry, acceleration/ deceleration ramps, I/ O offsets and scaling.
- Integrated each drive into the AS-i network.
- Configured local and remote control modes, ensuring safe handover between operator panels and SCADA control.
- Performed functional testing under load for conveyors, pumps, agitators, to validate torque, frequency and feedback performance.

3. AS-Interface (AS-i) Bus Network Commissioning

The field instrumentation was largely decentralized using an IFM AS-i Bus network for discrete and analog devices.

- Commissioned the AS-i topology, segmenting the network per process area to ensure isolation and fault containment.
- Integrated valves, limit switches, scales, proximity sensors and instruments via IFM AC1402 gateways and AS-i modules to the Siemens PLC.
- Configured addressing, diagnostics, and slave mapping within TIA Portal, ensuring robust data ntegrity and fast cyclic communication.
- Verified redundancy and fault diagnostics, ensuring that device-level interruptions did not propagate across process segments.

4. SCADA Development and System Integration

Developed and deployed a WinCC Professional SCADA system for plant-wide monitoring and control.

- Configured Engineering Station (ES) and Operator Station (OS) architecture with user access management, automatic backup policies, and redundant data logging.
- Developed over 20 process screens encompassing analog/ digital process displays, alarm windows, motor and valve faceplates, and operator dialogs.
- Implemented dynamic graphics, interlock visualization, trend pages, and historical data archiving for process analysis and fault tracking.

- Configured and optimized tagging, area hierarchy, and alarm management according to ISA-18.2 guidelines.
- Integrated Profinet data channels to external Microsoft SQL Server databases using Siemens Industrial Data Bridge, enabling external process data logging, historian replication, and downstream reporting for engineering and management teams.

5. Instrumentation, Valves, and Field Devices

Commissioned all analog and digital field instruments across multiple process areas.

- Configured and scaled analog devices including pressure transmitters, flow meters, pH sensors, level transmitters, and temperature sensors.
- Calibrated control and isolation valves, establishing correct PID loop direction and feedback.
- Developed and tested control logic for digital valves (open/ close feedback with interlocks) and analog valves (modulating valves for flow and level control loops).
- Validated instrument diagnostics, ensuring that all transmitters and field devices correctly reported status and faults to the control system.

6. Process Areas Covered

The automation scope covered the entire pilot plant process chain:

- Ore Scrubbing Section
- Pre-Thickening Area
- Acid Atmospheric Leach Area
- Resin-in-Pulp (RIP) Tails and Handling Sections
- Elution Circuit
- Uranium Precipitation Area
- Raw and Process Water Systems
- Water Treatment and Potable Water Plants
- Tails/ Scats Transfer Systems
- Excess Water Treatment Area
- Sulphuric Acid Dosing Area
- Hydrogen Peroxide and Sodium Hydroxide Dosing Areas
- Lime Handling System
- Compressed Air Plant

Each area was engineered for full automated sequence control, ensuring safety interlocks, startup sequencing, and process fail-safes were properly validated.

Project Management & Documentation

- Managed punch-list tracking, recording and resolving commissioning issues identified during cold and hot testing phases.
- Maintained progress tracking logs, ensuring all activities met project deadlines and compliance standards.
- Produced as-built documentation including updated PLC I/ O lists, network diagrams, electrical schematics and system architecture drawings.
- Delivered complete commissioning and handover documentation, including control narratives, loop check records, and system operation manuals.

Professional Summary

In this project, Mechatrolix executed complete automation lifecycle management—from PLC and SCADA engineering through to commissioning and documentation—for Tanzania's first uranium plant. The role demanded deep technical knowledge of Siemens TIA Portal, WinCC Professional, Profinet/ AS-i communications, and VSD integration within a high-integrity process control environment.

WinCC Professional <> Odoo Interfacing

Led the full-cycle design and implementation of a bidirectional integration between Siemens WinCC RT Professional and Odoo ERP for automated batch production and recipe management. The project spanned 11 weeks across four structured phases: system discovery and architecture design, inbound integration, outbound integration, and deployment. Conducted a comprehensive assessment of the existing TIA Portal PLC programs, WinCC configuration, and SQL Server infrastructure before designing the integration architecture. Developed inbound integration enabling automatic transmission of production orders and recipes from Odoo to the Siemens environment, including operator validation workflows and role-based access control on WinCC HMI screens. Implemented outbound integration capturing real-time weighing data, material consumption, and batch genealogy, with automated inventory reconciliation posted back to Odoo. Developed variance analysis, traceability reporting, and error handling with retry and buffering logic to ensure data integrity during network interruptions. Delivered full documentation covering system architecture, data flow mapping, operational procedures, and maintenance guides, along with operator and administrator training. The solution was deployed to production with a shadow operation period to validate accuracy before full cutover.

Power Station Lime Plant Refurbishment

PLC and HMI Engineering for the refurbished North and South Lime Plants at Kriel power station. This was a brown field project and Mechatrolix was required to develop a program and HMI screens for the new control philosophy with freedom to optimize when desired. The electrical and mechanical infrastructure required maintenance and a new FAT and SAT which Mechatrolix developed the project plan for.

Power Station Lime Plant Refurbishment

Contracted to specify an electrical, control and instrumentation solution for the Blinkpan Wasterwater Treament plant that's being overhauled. I have designed the MCC and PLC control panel and interfaced power quality measuring system and field instrumentation network using EPLAN and Pro Panel, the control system will run on a Siemens S7-1500 PLC with ET200 distributed I/ O stations. Cybersecurity and network hardening are within the scope and SCADA will be developed on WinCC Unified.